A Zeta-Pratt Fractal as a Three-Dimensional Volume

Each infinitesimal spherical shell is a different zoom level.

Elaboration of the proposed model | 5 October 2026

1 The Spatial Field

The observer is located at the fixed center $\boldsymbol{o} \in \mathbb{R}^3$. The image fills space: the same planar Zeta-Pratt pattern lies on every spherical shell, but with a zoom that depends on the radius. Greater depth can correspond simultaneously to greater magnification and lower brightness.

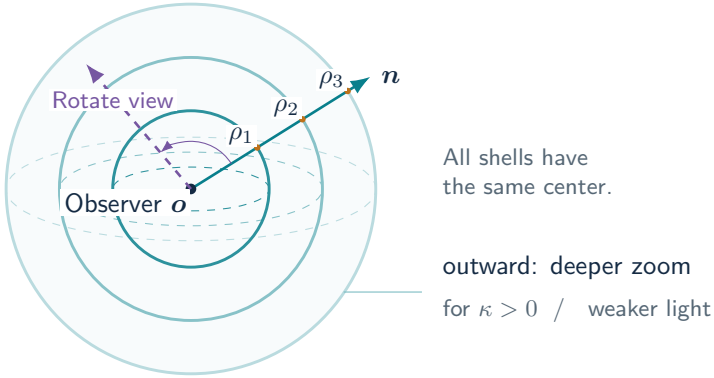


Figure 1: Symbolic cross-section through the volume. The drawn shells represent a continuous spectrum of radii.

For $\boldsymbol{x} \neq \boldsymbol{o}$, set

$$\rho = \|\boldsymbol{x} - \boldsymbol{o}\|, \qquad \boldsymbol{n} = \frac{\boldsymbol{x} - \boldsymbol{o}}{\rho} \in \mathbb{S}^2.$$

Basic formula: direction becomes a complex coordinate, depth becomes zoom.

$$C_{\boldsymbol{o}}(\boldsymbol{x}) = a(\rho)\, \mathcal{F}\left(z_* + \frac{\sigma(\boldsymbol{n})}{Z(\rho)} \right). \tag{1.1}$$

$\mathcal{F}(z)$	RGB color of the planar Zeta-Pratt pattern, componentwise in $[0, 1]$.
$z_* \in \mathbb{C}$	Fixed target point about which the pattern is magnified.
$\sigma(\boldsymbol{n})$	Stereographic coordinate of the direction; see Section 3.
$Z(\rho) > 0,\ a(\rho) \geq 0$	Zoom factor and brightness factor of the shell.

2 The Zeta-Pratt Texture

The Given Family

For positive integer indices, the recursive definitions are

$$f_1(z) = 1, \qquad f_2(z) = z, \qquad f_p(z) = 1 + f_{p-1}(z) \quad (p > 2 \text{ prime}), \tag{2.1}$$

and for composite n, the prime factorization

$$f_n(z) = \prod_{p|n} f_p(z)^{v_p(n)}. \tag{2.2}$$

Here $v_p(n)$ denotes the exponent of p in n. The requested extensions are $f_0(z) = 0$ and $f_{-n}(z) = -f_n(z)$. For a real index r and a complex index $s = a + ib$, set

$$f_r(z) = (1-t)f_m(z) + tf_{m+1}(z), \qquad m = \lfloor r \rfloor, \quad t = r - m, \tag{2.3}$$

$$f_s(z) = f_a(z) + if_b(z). \tag{2.4}$$

For fixed s, this yields a polynomial in z; moreover, $f_s(2) = s$. The index r must be distinguished here from the spatial depth ρ.

The Iterated Map of the Existing Image

We use the composition $\zeta \circ f_s$. In particular, from $f_4(z) = z^2$, $f_5(z) = z^2 + 1$, and $f_{1/2}(z) = 1/2$:

$$H(z) = \zeta(f_{4.5+0.5i}(z)) = \zeta\left(z^2 + \tfrac{1}{2} + \tfrac{i}{2}\right), \qquad z_{j+1} = H(z_j). \tag{2.5}$$

The zeta function is understood via its meromorphic continuation. Its Dirichlet series is valid for $\Re w > 1$; at $w = 1$ it has a simple pole [1]. Thus H has poles at $z^2 = (1-i)/2$.

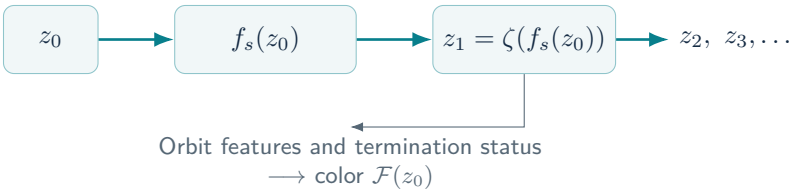


Figure 2: An initial coordinate is iterated and then colored. The image function $\mathcal{F}$ contains the chosen numerical representation.

What the Color Means

In practice, evaluation stops after at most N steps:

$$\mathcal{F}(z_0) = \text{Palette}(\text{status}, \text{step count}, \text{return test}, \arg z_j, \log(1 + |z_j|)).$$

The existing image uses such finite orbit features and numerical termination thresholds. It is therefore an *orbit representation*; the colors alone prove neither convergence nor membership in a mathematically exact Julia set. For the shell model, a fixed, bounded, and measurable image function $\mathcal{F}$ is sufficient.

3 From the Complex Plane to Every Spherical Shell

The Riemann sphere is the complex plane together with the added point ∞. It provides a natural directional coordinate. For $\boldsymbol{n} = (n_x, n_y, n_z) \in \mathbb{S}^2$, we use

$$\sigma(\boldsymbol{n}) = \frac{n_x + i n_y}{1 - n_z}, \qquad \boldsymbol{n} \neq \boldsymbol{N} := (0, 0, 1). \tag{3.1}$$

The south pole $(0, 0, -1)$ corresponds to $z = 0$, and the north pole $\boldsymbol{N}$ to ∞. This sign convention matches the sphere mapping used so far.

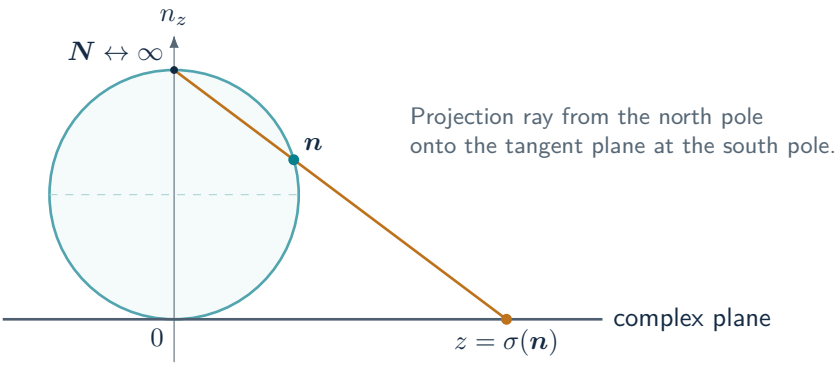


Figure 3: Stereographic projection in a two-dimensional cross-section. For the formula, the plane is coordinatized at half the geometric scale.

The inverse map for $z = u + iv$ is

$$\sigma^{-1}(z) = \frac{(2u,\, 2v,\, u^2 + v^2 - 1)}{1 + u^2 + v^2}. \tag{3.2}$$

Direct substitution gives $\|\sigma^{-1}(z)\| = 1$ and $\sigma(\sigma^{-1}(z)) = z$. On a shell of radius ρ, this directional vector becomes $\boldsymbol{x} = \boldsymbol{o} + \rho\boldsymbol{n}$.

The Texture of a Single Shell

We write

$$T_\rho(\boldsymbol{n}) = \mathcal{F}(A_\rho(\sigma(\boldsymbol{n}))), \qquad A_\rho(z) = z_* + \frac{z}{Z(\rho)}. \tag{3.3}$$

A_ρ is an affine Möbius transformation of the Riemann sphere [2]. It shifts the target of the excerpt and reduces the sampled distances in the plane. The actual spherical shell is a copy of the direction sphere; the spatial radius controls the zoom through $Z(\rho)$.

Projection Pole and Center

The coordinate chart has a pole at the north pole. A second chart $w = 1/z = (n_x - i n_y)/(1 + n_z)$ covers its neighborhood, but provides *no* automatic continuous extension of the fractal color to $z = \infty$. For an image defined everywhere, a fixed color can be assigned to the single projection pole. Likewise, one may set $C_{\boldsymbol{o}}(\boldsymbol{o}) = 0$; the direction is undefined there. Both are rendering rules.

4 Depth as a Logarithmic Zoom Coordinate

A simple model choice is

$$Z(\rho) = Z_0 e^{\kappa\rho}, \qquad Z_0 > 0, \quad \kappa \in \mathbb{R}. \tag{4.1}$$

Then

$$\log Z(\rho) = \log Z_0 + \kappa\rho, \qquad \frac{Z(\rho+\Delta\rho)}{Z(\rho)} = e^{\kappa\Delta\rho}. \tag{4.2}$$

Shells of equal thickness therefore correspond to equal *multiplicative* zoom steps. For $\kappa > 0$, moving outward zooms in; for $\kappa < 0$, it zooms out. For $\kappa = 0$, all shells carry the same texture.

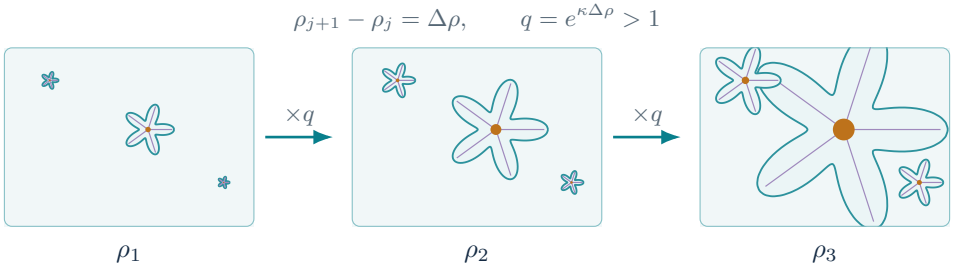


Figure 4: Symbolic magnification sequence. The flower symbols illustrate only the zoom idea; they are not computed Zeta-Pratt excerpts.

What Happens Along a Ray

For a fixed direction $\boldsymbol{n}$, the initial coordinate of the fractal iteration is

$$z(\rho, \boldsymbol{n}) = z_* + Z_0^{-1} e^{-\kappa\rho}\sigma(\boldsymbol{n}), \qquad \frac{\partial z}{\partial \rho} = -\kappa(z - z_*). \tag{4.3}$$

For $\kappa > 0$, it approaches the target point z_* exponentially as depth increases. Colors and orbit features can change strongly in the process. The zoom construction itself does not guarantee exact self-similarity of the pattern.

Zoom over a Given Time

Let τ be the interaction time and ν the desired logarithmic zoom rate. If the camera preferentially displays the depth $\rho_c(\tau)$, set, for $\kappa \neq 0$,

$$\rho_c(\tau) = \rho_c(0) + \frac{\nu}{\kappa}\tau, \qquad Z(\rho_c(\tau)) = Z(\rho_c(0))e^{\nu\tau}. \tag{4.4}$$

The center $\boldsymbol{o}$ remains fixed. A different depth of the same volume is selected. For a finite radial range, this motion ends at its boundary; a depth selection is described in Section 6.

5 The Camera at the Fixed Sphere Center

A rotation is described by a matrix $Q(\tau) \in \mathrm{SO}(3)$:

$$Q^{\mathsf{T}}Q = \mathrm{Id}, \qquad \det Q = 1.$$

For image coordinates (u, v) and a positive camera factor $M(\tau)$, define

$$\boldsymbol{n}_{\mathrm{cam}}(u, v; M) = \frac{(u/M,\, v/M,\, 1)}{\sqrt{1 + (u^2 + v^2)/M^2}}, \qquad \boldsymbol{n} = Q(\tau)\boldsymbol{n}_{\mathrm{cam}}. \tag{5.1}$$

Larger M reduces the field angle for an unchanged image area. The corresponding ray always passes through the same point:

$$\boldsymbol{x}(\rho; u, v, \tau) = \boldsymbol{o} + \rho\, Q(\tau)\boldsymbol{n}_{\mathrm{cam}}(u, v; M(\tau)). \tag{5.2}$$

For an initial view toward the south pole, for example, $Q(0) = \mathrm{diag}(1, -1, -1)$ is suitable.

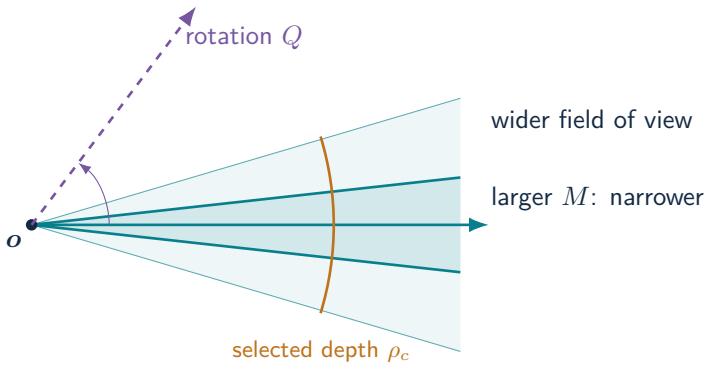


Figure 5: View rotation and angular zoom change the ray directions. Depth selection changes which shells are preferentially visible.

Control	Effect	Parameter
Rotate	Different viewing direction, same center.	$Q(\tau)$
Angular zoom	Narrower section of the spherical panorama.	$M(\tau)$
Depth zoom	A different zoom level of the fractal is emphasized.	$\rho_c(\tau)$

The Volume Remains Fixed

As long as z_*, $Z(\rho)$, $a(\rho)$, and $\mathcal{F}$ remain unchanged, the same field is being viewed. Only the camera and its depth weighting change. The rotation is therefore *not* a motion of the sphere center.

The angular zoom M and the fractal zoom Z are different quantities. They may be coupled, but need not be. By contrast, changing the target point z_* or rotating the texture in formula (1.1) changes the constructed field. For the desired look inward, a fixed texture with a movable camera and movable depth selection is sufficient at first.

6 Attenuated Light and Transparent Shells

For the desired attenuation, choose a positive length scale ℓ and

$$a(\rho) = e^{-\rho/\ell}. \tag{6.1}$$

At $\rho = \ell$, the remaining fraction is $e^{-1} \approx 0.368$. This is a prescribed image weighting. The additional relation $\Delta t = \rho/c$ may interpret depth as light-travel time in a static model; it does not follow from the fractal zoom.

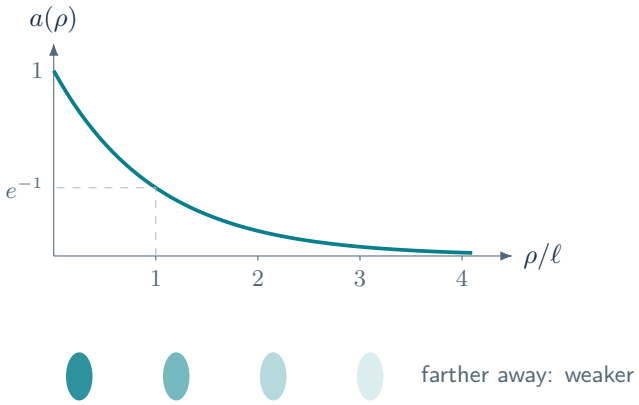


Figure 6: Exponential attenuation with depth. The points illustrate only relative brightness.

An Image from the Entire Volume

For a simple transparent, additively luminous volume, define

$$I(\boldsymbol{n};\tau) = \int_0^{\rho_{\max}} W(\rho;\tau)\, a(\rho)\, \mathcal{F}\!\left(z_* + \frac{\sigma(\boldsymbol{n})}{Z(\rho)}\right) \mathrm{d}\rho. \tag{6.2}$$

Integration is performed along a ray, not over the area of an entire spherical shell. Therefore no area factor $4\pi\rho^2$ appears here. The weighting W determines how strongly a given depth contributes to the image. $W = 1$ sums all shells; a normalized narrow window around $\rho_c(\tau)$ preferentially shows one zoom level. If the texture is continuous there, then in the limit of an increasingly narrow window, $I \to a(\rho_c) T_{\rho_c}(\boldsymbol{n})$.

Numerical Representation and Brightness Bound

A discrete representation with shell spacings $\Delta\rho_j$ is

$$I(\boldsymbol{n};\tau) \approx \sum_j W(\rho_j;\tau) a(\rho_j) T_{\rho_j}(\boldsymbol{n})\, \Delta\rho_j. \tag{6.3}$$

The factors $\Delta\rho_j$ prevent the image from becoming brighter merely because more shells are sampled. For $W = 1$ and every color component k,

$$0 \le I_k \le \int_0^{\rho_{\max}} e^{-\rho/\ell}\, \mathrm{d}\rho = \ell(1 - e^{-\rho_{\max}/\ell}) \le \ell.$$

A fixed exposure and color mapping are then applied for display. A narrow depth weighting also prevents many zoom levels from blending simultaneously into a low-contrast average.

7 What Does It Mean That Every Point Is an Observation Center?

The Same Construction Rule Around Every Center

For every $\boldsymbol{o} \in \mathbb{R}^3$, formula (1.1) defines an admissible model. If the target point, orientation, and zoom law remain the same, then for every translation $\boldsymbol{b}$,

$$C_{\boldsymbol{o}+\boldsymbol{b}}(\boldsymbol{x}+\boldsymbol{b}) = C_{\boldsymbol{o}}(\boldsymbol{x}). \tag{7.1}$$

That is: *the family of models is translation-covariant.* Every point can be the center of a shell volume constructed by the same rule. Under such recentering, the field changes in the fixed world coordinate system.

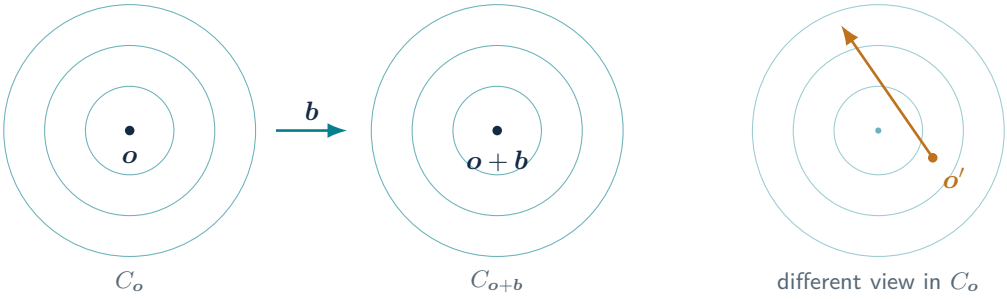


Figure 7: Left and center: the entire model is translated. Right: another observer looks into the original, fixed volume.

Another Observer in a Fixed Volume

If the world field $C_{\boldsymbol{o}}$ remains fixed and the observer is at $\boldsymbol{o}'$, then the ray $\boldsymbol{x}(u) = \boldsymbol{o}' + u\boldsymbol{n}'$ passes through different shell intersections:

$$\rho(u) = \|\boldsymbol{o}' - \boldsymbol{o} + u\boldsymbol{n}'\|, \qquad \boldsymbol{n}(u) = \frac{\boldsymbol{o}' - \boldsymbol{o} + u\boldsymbol{n}'}{\rho(u)}. \tag{7.2}$$

In general, $\rho(u) \neq u$, and $\boldsymbol{n}(u)$ is not constant. The new observer therefore does not see an identical concentric zoom sequence.

Exact Equality and Statistical Homogeneity

A fixed field satisfying $C(\boldsymbol{x}+\boldsymbol{b}) = C(\boldsymbol{x})$ for *all* translations $\boldsymbol{b}$ would have to be constant: with $\boldsymbol{b} = \boldsymbol{y} - \boldsymbol{x}$, one immediately gets $C(\boldsymbol{y}) = C(\boldsymbol{x})$. A nonconstant shell fractal cannot satisfy this strong pointwise condition. Statistical homogeneity, by contrast, means a translation-invariant probability law; it would be an additional model requirement and does not guarantee identical images.

The proposed model is a **three-dimensional field of continuously zoomed spherical shells**. The center remains fixed when rotating and when selecting other depths. The same construction rule can be applied around any center; this describes a family of volumes.

Sources and model status. The Pratt definitions and the shell idea come from the conversation; the iteration corresponds to the supplied renderer `render.cpp`. The zoom law, attenuation, and image integration are explicit model choices here.

[1] NIST, *DLMF*, Section 25.2: https://dlmf.nist.gov/25.2.

[2] J. C. Baez, *The Octonions*, Section 3.3, Möbius transformations of the Riemann sphere: https://math.ucr.edu/home/baez/octonions/node11.html.